GNN-Based Multi-Omics Integration for Predicting Cancer Drug Response

MCS16

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Agenda

01 Problem & Opportunity

Aditya Ansh



02 Target Users & Project Breakdown

Wasif Hussain



03 Project Architecture

Charmaine Chee



04 Demo & What's Next

Shaahid Marikar





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Problem Statement

01

41 Days-11 Months

(Chehade et al., 2026)

Genomic testing delays faced by advanced cancer patients awaiting matched therapy decisions

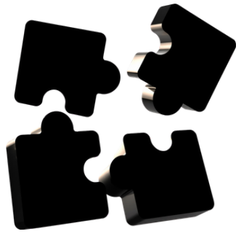


02

40%+ Patients

(Chen, 2026)

Advanced cancer patients missing genomic testing entirely due to fragmented data pipelines



03

“Data Tsunami”

(Aldea et al., 2023)

Oncologists struggle to interpret multi-omics variants without integrated bioinformatic support



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Why This Project Exists

fraction of physicians who accurately interpret complex genomic variants

14.6%

(PubMed, n.d.)

The bottleneck

31%

response rate with biomarker-matched therapies vs. 10.5% standard care.

higher response when bioinformatic support is provided

Opportunity

~3x

The Gap

(Schwaederle et al., 2015)

10.5%

response rate without matched therapy - nearly 3x lower outcomes

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Who is this for?



Cancer pharmacology researchers

Scan drug candidates against cancer cell line profiles before costly wet lab assays



Bioinformaticians

Benchmark our model's approach against their own model



Academic users & students

Explore how genomic features relate to drug sensitivity through our application's dashboard

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What does the system deliver

The gap we are trying to bridge

Not every drug works on every cancer, and lab testing to find out is slow

You pick

- A cancer · 532 cell lines
- A drug · 498 compounds



Our model

Answers in seconds

You get

- Sensitive or resistant
- A confidence score
- Graphs that give context

WASIF HUSSAIN

The black box - How it works, and how we know it works

1. What powers it

- **Cancer's genetic fingerprint**
- **Drug's chemical structure**
- **Thousands of lab results**

2. What's new

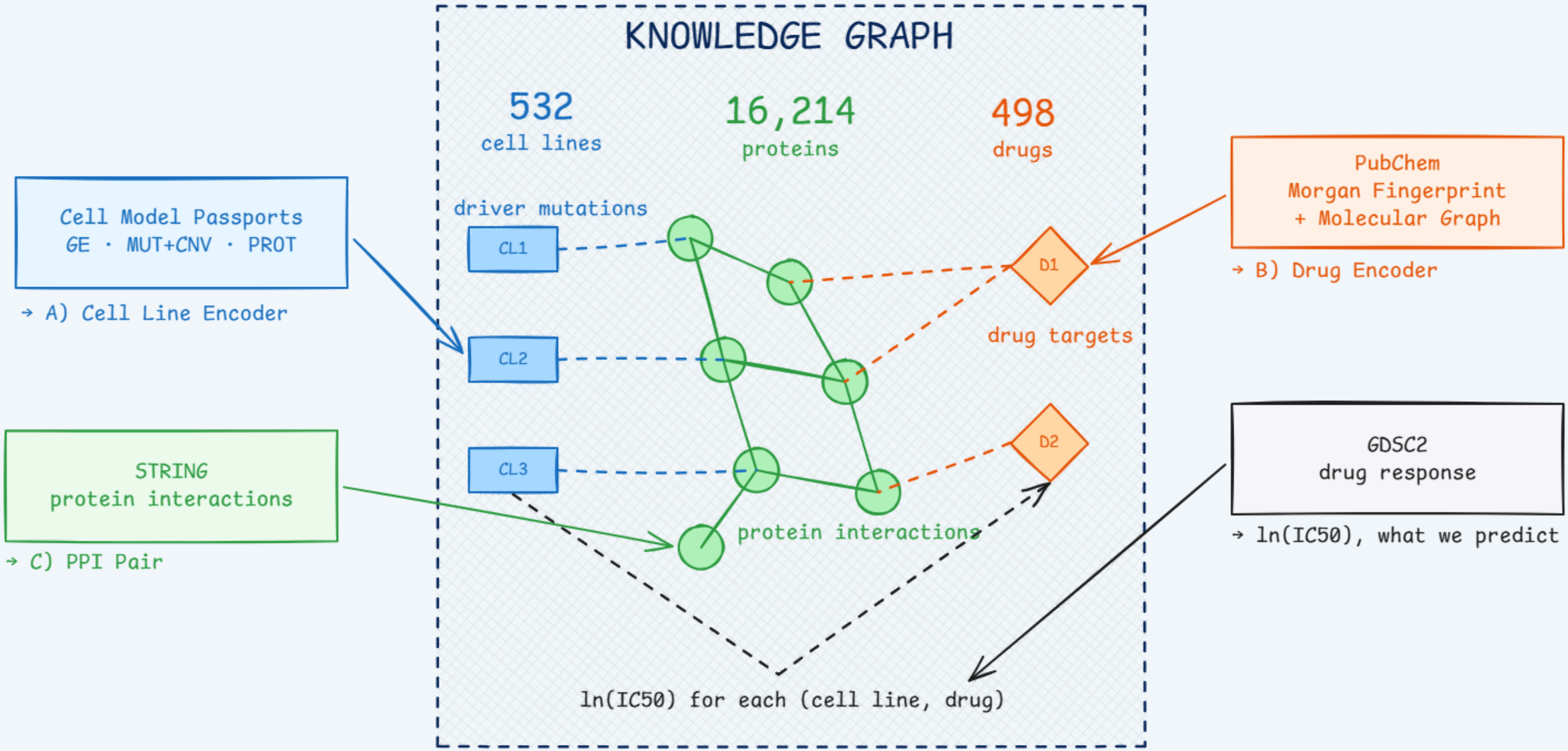
Same public lab data as other researchers, with new additions

- Drug read as atom network
- Genetic layers fused anew

3. How we proved it

- Comes close to standard model
- Every part improves prediction
- Tested on unseen cancers

From Public Data to a Biological Graph



Our Model

One switch per addition

1. MoGraphDRP: published base, reproduced



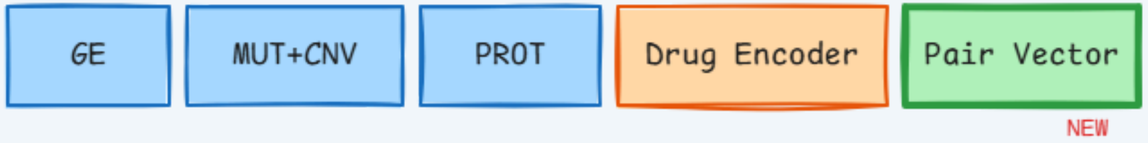
switch on

2. + Proteomics: third omics input



switch on

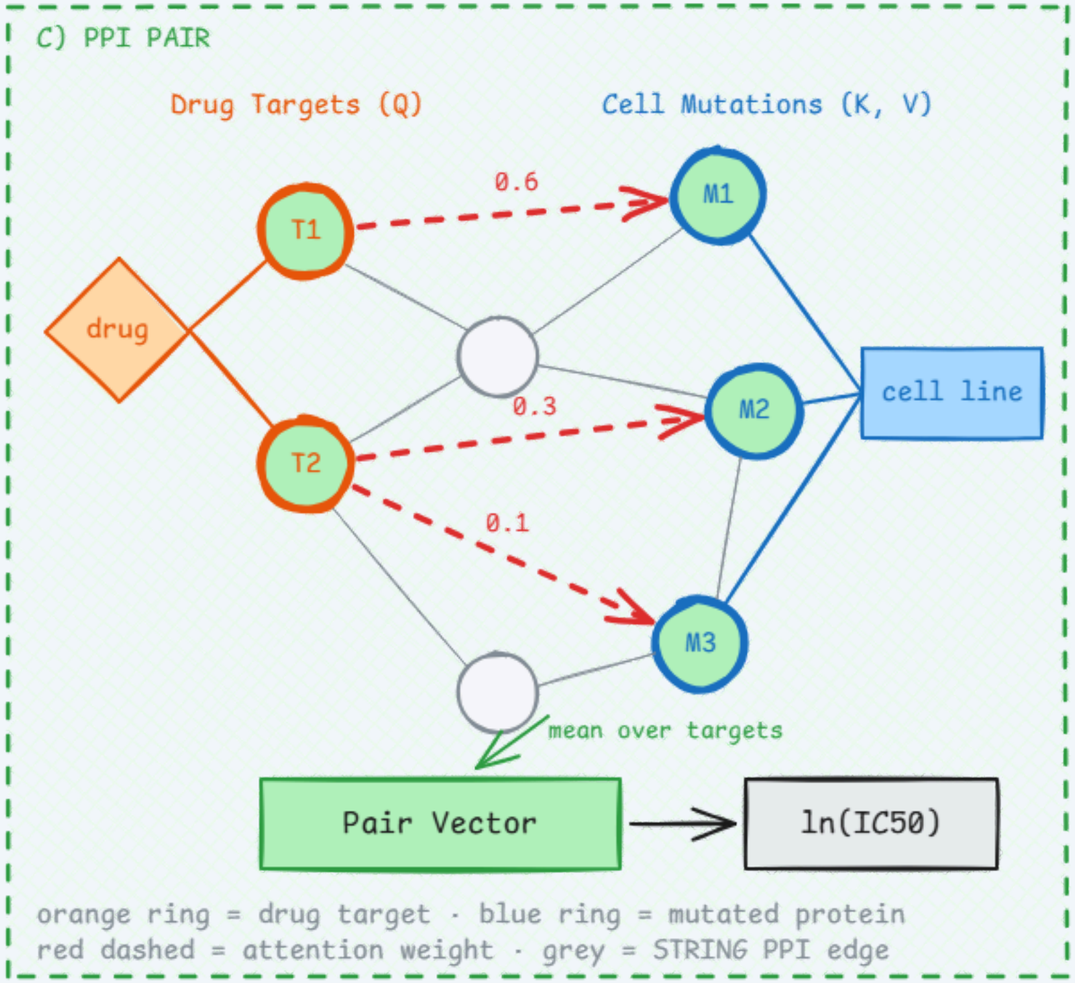
3. + Cross-Attention Fusion: full model



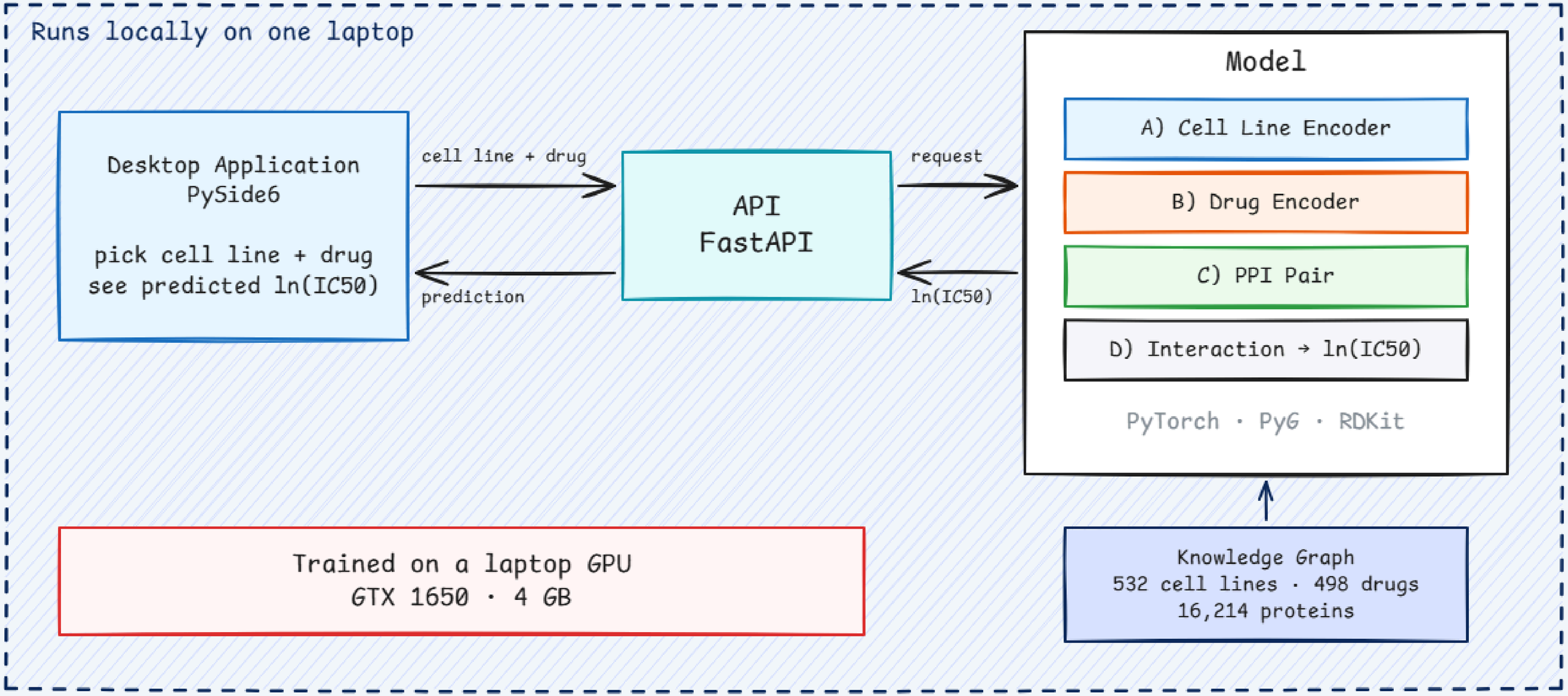
Tested on unseen cell lines



THE NOVELTY: Cross-Attention Fusion



The Application



FINAL ARCHITECTURE

multi-omics + GNN model for predicting drug response, ln(IC50)

CHARMAINE CHEE

A) CELL LINE ENCODER

Gene expression (GE)
128 PCA components

Mutation + CNV
128 PCA components

ADDED a
Proteomics
128 PCA components

MLP Branch
128 → 256 → 128

MLP Branch
128 → 256 → 128

MLP Branch
128 → 256 → 128

CONCAT
GE
MUT
PROT

Cell Vector
384-d
GE MUT PROT

B) DRUG ENCODER

Morgan Fingerprint
1 0 1 1 0 0 1 0
2048 bits

Fingerprint MLP
2048 → 1024 → 512 → 256 → 128

ADDED b
Molecular Graph
atoms and bonds

GNN
GCN × 3, sum pooling
1024 → 128

Gated Sum
g
+
gated blend

Drug Vector
128-d

C) PPI PAIR FEATURES

STRING PPI Graph
16,214 proteins

Drug Targets
known proteins

Cell Mutations
driver mutations

graph lookup at a glance
target is mutated
T
T
M
M
1 hop
T drug target
M mutated in this cell line

ADDED c
Pair Features
5 numbers per pair
1 target mutated? 0/1
2 hop proximity
3 mutations 1 hop away
4 has target? 0/1
5 has mutation? 0/1

D) PREDICTION, WITH PER-DRUG TARGET SCALING

CONCAT
cell 384
drug 128
pair 5
517-d

final model = base + four added parts
ADDED a proteomics branch
ADDED b molecular-graph GCN GNN
ADDED c PPI pair features
ADDED d per-drug target scaling

ADDED d
Drug Statistics
mean, std per drug

looked up by drug

Predictor MLP
517 → 512 → 128 → 1

z

De-standardise
z × std + mean

MSE loss, training only

Training Target
z = (y - mean) / std

ln(IC50)
predicted drug response

legend
cell line data
drug data
PPI / pair data
learned module
GNN graph neural net
target scaling
added to base

what IC50 means cell viability
resistant
50%
sensitive
IC50
log [drug]

Requirements

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Req	Description	Type	Status
R01	Ranked drug efficacy list + confidence intervals	FR	✓
R02	Delete uploaded files after session	NFR	✓
R03	Real-time progress feedback	NFR	✓
R04	Molecular profile summary, top 5 mutations	FR	✓
R05	Export results to PDF/CSV	FR	✓
R06	Multi-omics + SMILES processing, via cross-attention fusion	FR	✓
R07	Runs within 4-8GB VRAM	FR	✓

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Limitations & Outcome Discussion

Planned, not achieved	Why
Integrate a component PPI over our GNN	**Did not offer enough useful connections to help predictions**
Full SHAP interpretability - SHAP explains reasoning behind prediction	**Priority went to modeling; minimal check only**

Current Limitations

Weak prediction on new drugs
needs drug-target data we don't always have

Narrow interpretability
we only explain direct target-mutation matches

LIVE DEMO

DEMO

Desktop dashboard walkthrough

01 Dataset Upload

02 Model Execution Log

03 Final Results





SHAAHID MARIKAR

What's Next?

Improve Model Performance & Usability

01 **Fix & Retest PPI Attention**

test different versions of PPI Attention that could pottentiall improve the model

02 **Model Growth Rate**

Diagnosed covariate, cuts remaining error 26%

CONTINUE: GNN + PPI

Generalization & Interpretability

Sparse drug-target coverage caps unseen-drug accuracy; full SHAP module remains open

Final Model Test RMSE:

- Grouped split: 1.27 vs. 1.33 baseline
- Random split (reference): 0.96 vs. 1.02 baseline

Thank You!

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